PHARMACEUTICALS IN RIVERS: A SILENT AQUATIC CRISIS

Dinethya Senojee Perera (s224771638)

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Executive Summary

Pharmaceutical residues, particularly antidepressants and antiepileptic drugs (AEDs), are increasingly detected in freshwater rivers as a consequence of human medicine use, excretion, incorrect disposal and inadequate wastewater treatment. Conventional wastewater treatment plants demonstrate poor removal efficiency for key compounds including carbamazepine (Antiepileptic drug) and venlafaxine (Antidepressant drug), resulting in continuous discharge into freshwater systems. At environmentally relevant concentrations, these residues induce neurobehavioral impairment, reproductive toxicity, growth inhibition and bioaccumulation across multiple aquatic species, threatening the ecological integrity of freshwater ecosystems. Without targeted intervention across source control, treatment infrastructure and regulatory frameworks, pharmaceutical contamination will continue to pose a significant and escalating risk to Australian freshwater environments and the ecosystems they sustain.

Overview

The presence of pharmaceuticals and their metabolites in the aquatic environment depends mainly on the frequency and amount of medicines used. The content of individual antidepressants and their metabolites in aquatic ecosystems depends on the region of the world, as well as on the popularity, and the extent to which these compounds are removed before treated effluent enters freshwater systems (wastewater treatment efficiency). Problems with the effectiveness of the removal of pharmaceuticals depend on the different levels of polarity and electrostatic activity (Zdziobek and Stanek, 2020).

Pharmaceuticals enter wastewater treatment plants (WWTPs) by wastewater from the disposal of unused or expired drugs in toilets as well as human excretion and hospital effluents (Salahinejad et al., 2023). Human excretion is considered to be the primary source of the pharmaceuticals in the environment (Giebułtowicz and Nałęcz-Jawecki, 2014).

Antidepressants and AEDs (Antiepileptic Drugs) are frequently prescribed pharmaceuticals. Even though antidepressants are excreted mainly as metabolites (biologically active or not), part of the administrated dose remains unchanged. Conventional activated sludge treatments reveal only moderate potential to remove antidepressants (Giebułtowicz and Nałęcz-Jawecki, 2014).

AEDs and their metabolites/transformation products were widely found in surface water, groundwater and drinking water (Liu et al., 2023). AEDs in the environment requires refined analytical techniques and, in most cases, pre-treatment procedures, such as liquid-liquid extraction and solid-phase extraction (Klanovicz et al., 2021).

Even at low concentrations (micrograms/nanograms per litre), these drugs can reveal some adverse effect on aquatic life (Liu et al., 2023).

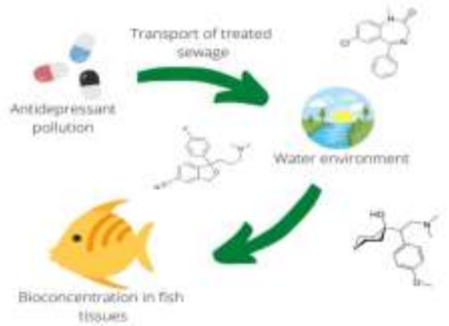


Fig 1: Route of spread of antidepressants in the aquatic environment (Zdziobek and Stanek, 2020)

Almost all AED drugs are detectable in natural aquatic ecosystems, but diazepam (DZP) and carbamazepine (CBZ) are among the most widely detected AEDs to date (Salahinejad et al., 2023). Among antidepressants detected in aquatic environments, fluoxetine, sertraline, paroxetine and venlafaxine are important compounds of concern. Fluoxetine and sertraline are widely studied selective serotonin reuptake inhibitors (Giebułtowicz and Nałęcz-Jawecki, 2014), while venlafaxine and its metabolite O-desmethylvenlafaxine are commonly reported in wastewater-impacted waters (Zdziobek and Stanek, 2020). Paroxetine is also relevant because its lipophilic nature may influence how it is removed during wastewater treatment or adsorbed onto sludge and organic matter (Zdziobek and Stanek, 2020). .

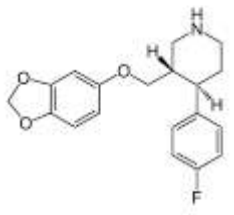


Fig 2: Structure of Paroxetine

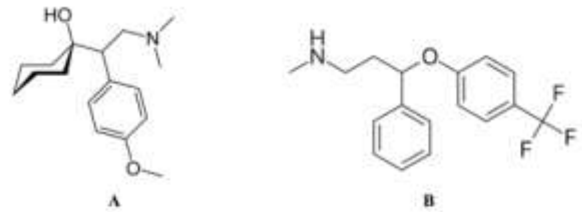


Fig 3: Structure of venlafaxine (A) and fluoxetine (B)

Continuous exposure of aquatic organisms to antidepressants, AEDs and their metabolites is a toxicological concern because these drugs have a substantial capacity to induce neurotoxicity and behavioural abnormality in non-target species (Salahinejad et al., 2023). In freshwater fish, antidepressant residues may affect brain-mediated behaviours such as feeding, movement, predator avoidance and escape responses. These effects are important because impaired food detection, reduced mobility or slower responses to predators can decrease survival even when the exposure does not cause immediate death (Zdziobek and Stanek, 2020). Antidepressant and AED exposure may also influence growth, reproduction and cellular stress responses, suggesting that long-term contamination could affect both individual organisms and freshwater populations (Klanovicz et al., 2021).

Pharmaceuticals' half-life ($t_{1/2}$) on environmental matrices depends significantly on the pharmaceutical compound itself and on the medium, reaching up to 120 h. This parameter helps to estimate how long the pollutant will be present in the environment. Solubility is also an important physical-chemical property, indicating whether the contaminant has more affinity with water or with organic substrates. It is directly related to the distribution coefficient and the octanol-water partition coefficient (log K_{ow}). The pK_a is also relevant in environmental samples, which are usually acidified before the extractions and analysis of pharmaceuticals. In fact, the extractability is reduced for compounds exhibiting high pK_a values (Klanovicz et al., 2021). Proportion of the parent drug excreted unchanged, and the formation of metabolites is important because both the parent drug and its metabolites may enter wastewater systems and contribute to freshwater contamination (Zdziobek and Stanek, 2020).

Methodology

The toxicological risk of antidepressant and anti-epileptic residues in freshwater rivers should be investigated using an integrated approach combining chemical monitoring, ecotoxicological testing and risk assessment.

Chemical monitoring would identify whether pharmaceutical residues are present in freshwater systems, while toxicity testing would determine whether measured concentrations are likely to cause adverse effects in aquatic organisms.

Water samples should be collected from multiple sites to determine the source and distribution of contamination. Sampling locations should include an upstream control site, wastewater treatment plant influent and effluent, the effluent discharge point, and several downstream river sites. Sediment and aquatic organism samples may also be collected where possible, as some compounds may adsorb to organic matter or accumulate in biological tissues (Giebułtowicz and Nałęcz-Jawecki, 2014).

What is measured	**Why it is measured**
Upstream river water	Provides a background/control concentration before wastewater enters the river.
Wastewater influent	Shows the amount of pharmaceuticals entering the wastewater treatment plant from human use, excretion and medicine disposal.
Wastewater effluent	Shows how much pharmaceutical residue remains after treatment.
Effluent discharge point	Identifies the direct concentration entering the freshwater river from treated wastewater.
Downstream river water	Shows how far the contaminants travel and whether concentrations decrease due to dilution, degradation, sediment binding or biological uptake.
Sediment	Some pharmaceuticals may bind to organic matter or particles rather than stay dissolved in water. Helps detect compounds that may persist in the riverbed.
Aquatic organisms/tissues	Shows whether pharmaceuticals are bioavailable and can enter living organisms.

(Giebułtowicz and Nałęcz-Jawecki, 2014)

- If the upstream site has very low or no pharmaceuticals, but downstream sites have higher levels, that suggests the wastewater treatment plant is a major source.

Wastewater removal efficiency

Antidepressants:

Antidepressants differ in how effectively they are removed during treatment. More lipophilic compounds are more likely to adsorb onto sewage sludge, suspended particles and microorganisms, which can increase their removal from the water phase.

Drug	Removal Percentage
Paroxetine	~90%
Fluoxetine	~40–90%
Venlafaxine	~25%

(Zdziobek and Stanek, 2020)

AEDs:

Parameter to measure	Example data for DZP	Example data for CBZ
Persistence	Water half-life >365 days	Half-life of approximately 37 days; very slowly degraded by sunlight
WWTP removal efficiency	<50% removal	<10% removal
Observed concentration range (worldwide)	• Reported aquatic range from a few ng/L to 840,000 ng/L • Wastewater concentration - Median 0.016 ± 0.040 µg/L • Surface water range - <1–79.33 ng/L	• Wastewater median: 0.272 ± 0.658 µg/L • Surface water median: 0.059 ± 0.105 µg/L • Reported aquatic range from a few ng/L to 10,300 ng/L

(Salahinejad et al., 2023)

Proportion of the parent drug-metabolite excreted

AED Data:

25% of CBZ is excreted in urine after liver metabolization, as it's primary metabolite carbamazepine-10,11-epoxide, while 40–50% of PMD is excreted unchanged by urine (Klanovicz et al., 2021).

Antidepressant data:

Fluoxetine is excreted unchanged in urine approximately ~10%, Paroxetine ~3% and Sertraline ~0.2% (Giebułtowicz and Nałęcz-Jawecki, 2014) .

These data justify the need to monitor both unchanged parent compounds and their metabolites.

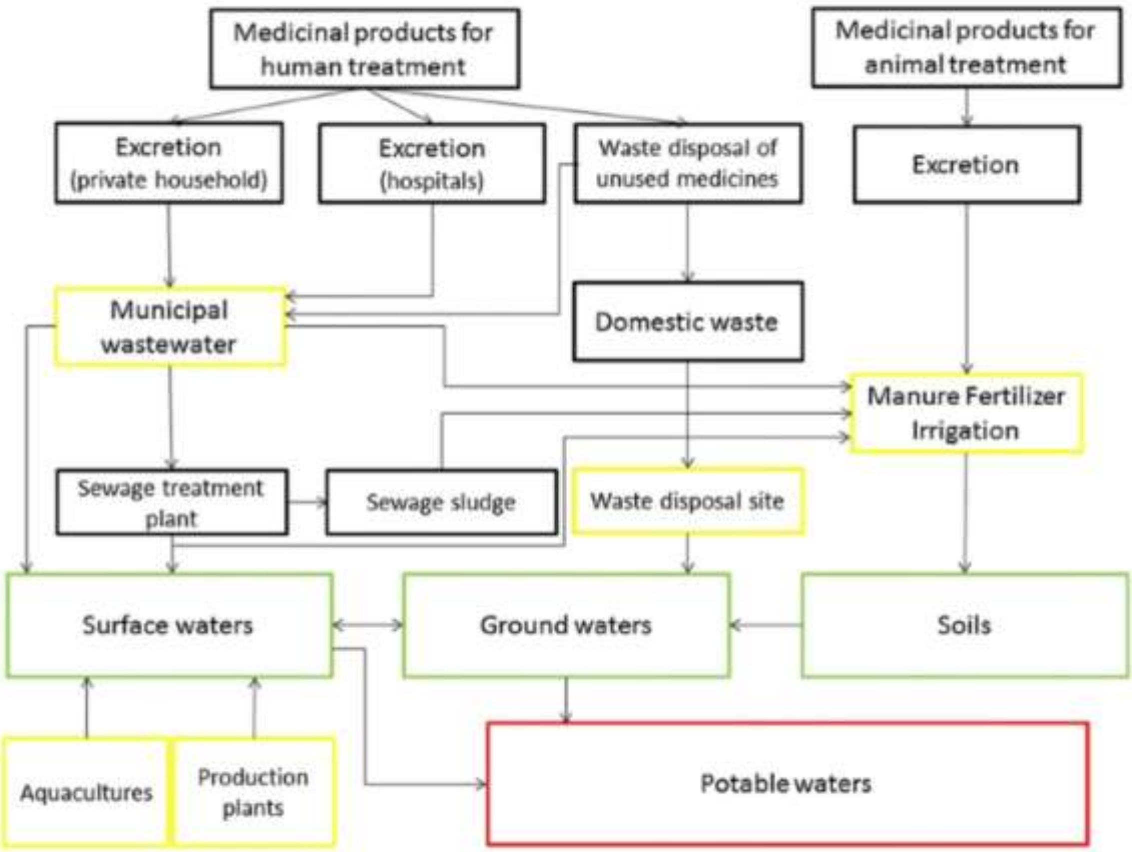


Fig 4: Intermediate steps encountered between the point source and the receiving environmental compartment (Bagnis et al., 2018)

Discussion

The toxicological risk of pharmaceutical residues in freshwater rivers is mainly associated with chronic exposure rather than acute poisoning.

Antidepressants	**Toxicity response observed**
SSRIs: sertraline, fluoxetine, paroxetine	Reduced reproduction in *Ceriodaphnia dubia* (water flea), with NOEC values of 9 µg/L for sertraline, 89 µg/L for fluoxetine and 220 µg/L for paroxetine.
Sertraline	Reproductive toxicity for *Daphnia magna* reported with a NOEC of 32 µg/L.
Fluoxetine	Reduced reproduction in freshwater mudsnail *Potamopyrgus antipodarum* at concentrations as low as 0.81 µg/L.
Fluoxetine and sertraline	Growth inhibition in *Pseudokirchneriella subcapitata* (green algae), with 48h EC_{50} values of 24 µg/L for fluoxetine and 12.1 µg/L for sertraline.
Fluoxetine	*Desmodesmus subspicatus* (green algae) was more sensitive to fluoxetine (EC_{50} of 2.1 µg/L).
Venlafaxine	Affected fathead minnow - predator avoidance at 0.5 µg/L and showed chronic toxicity to plants at 0.1 µg/L.

(Giebułtowicz and Nałęcz-Jawecki, 2014).

Anti-epileptic drugs	Toxicological analyses are mainly based on LD_{50} or LC_{50} indicators.
CBZ in *Daphnia magna*	Reduced feeding ability, decreased negative phototactic behaviour, oxidative stress.
Carbamazepine in algae	Decreased algal growth; at 50–100 mg/L (higher concentrations), reduced biochemical characteristics and photosynthetic activity.

(Klanovicz et al., 2021)

NOEC = No Observed Effect Concentration (the highest tested concentration where no harmful effect is observed)

EC_{50} = Half Maximal Effective Concentration

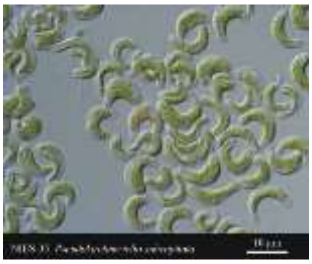
A continuous exposure of aquatic organisms to antidepressants and their metabolites causes specific adverse reactions. In fish, antidepressant exposure may affect growth, antioxidant defence systems and detoxification processes. These compounds may also increase lipid peroxidation (oxidative damage to lipids in cell membranes), contribute to liver damage and cause genotoxic effects. A major toxicological concern is neurobehavioral impairment, as disruption of brain-mediated functions may reduce the ability of fish to locate and consume food. Antidepressants may also affect reproductive processes, which could reduce population renewal over time. Also, exposure may impair escape responses during predator attacks (Zdziobek and Stanek, 2020).

The main bioaccumulation targets for both DZP and CBZ are brain and liver of fish. After multiple days of exposure, DZP levels in fish bodies can be up to 17 times higher than in the surrounding water. Across tissues, DZP and CBZ bioaccumulation reaches steady-state concentrations after 24 h (Salahinejad et al., 2023).

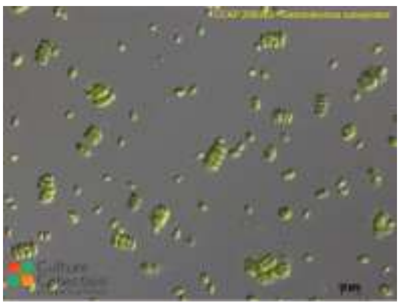
AEDs may disrupt key neurotransmitter systems in fish, including GABAergic signalling (inhibitory nerve signalling that reduces neuronal activity), glutamatergic signalling (excitatory nerve signalling that increases neuronal activity) and serotonergic signalling (serotonin-based signalling involved in behaviour, feeding, aggression and stress responses), as well as parasympathetic neurotransmission (rest and digest). In teleost (bony) fish, these effects may extend beyond the nervous system, with some AEDs also associated with estrogenic activity (disruption of estrogen-related hormone signalling that may affect reproduction and growth) and oxidative stress (cellular damage caused when reactive oxygen species exceed the antioxidant defences). These physiological changes can lead to dose-dependent effects on anxiety-related behaviour, locomotor activity, social behaviour, food uptake, learning and memory (Salahinejad et al., 2023).

Daphnia magna

Potamopyrgus antipodarum



Pseudokirchneriella subcapitata



Desmodesmus subspicatus

Ceriodaphnia dubia

Fathead minnow

Management Actions

Pharmaceutical residues enter rivers mainly through human medicine use, excretion, incorrect disposal and wastewater discharge, so management should focus on source control, monitoring and treatment improvement.

Official Australian sources support this direction: the Australian Government funds the National Return and Disposal of Unwanted Medicines Program for safe disposal of expired and unwanted medicines, and the TGA (Therapeutic Goods Administration) advises that incorrect medicine disposal can harm the environment and that unwanted medicines should be returned to community pharmacies (Therapeutic Goods Administration (TGA), 2022), (Australian Government Department of Health, Disability and Ageing, 2025).

Management Actions:

Introduce a pharmaceutical stewardship levy for high-risk medicines

The Department of Health should consider introducing a manufacturer-funded environmental levy on pharmaceuticals that are difficult to remove during wastewater treatment. The revenue from this levy could be used to support advanced wastewater treatment upgrades at high-risk locations. This approach would place some responsibility on pharmaceutical manufacturers, rather than relying only on water authorities or taxpayers to manage the environmental impacts of these medicines.

Establish Environmentally Weighted Prescribing Incentives Under the PBS

The Pharmaceutical Benefits Scheme (PBS) should encourage the use of medicines with lower environmental persistence when they are clinically equivalent to other treatment options. Where two medicines are similarly effective and safe for patients, the PBS could support the option that is less likely to persist in wastewater and freshwater systems.

Mandate Blister Pack and Dispensing Quantity Reform

The Department should regulate dispensers to issue smaller pack sizes for high-risk pharmaceuticals to reduce the volume of unused medicine in households — a primary source of incorrect disposal. Blister pack reforms that make individual tablets harder to hoard in bulk would structurally reduce the pool of medicines available for improper disposal.

Create a National Pharmaceutical Environmental Incident Register

The Department should establish a publicly accessible register where environmental agencies, councils and researchers can report freshwater pharmaceutical contamination events. This creates accountability, enables pattern recognition across jurisdictions, and generates the national evidence base currently absent in Australia — forcing regulatory response when contamination thresholds are repeatedly breached in the same catchment.

Fund Catchment-Specific Risk Zoning for Drinking Water Protection

The Department, in partnership with the National Health and Medical Research Council (NHMRC), should commission catchment-level pharmaceutical risk maps identifying river systems supplying drinking water that are downstream of major WWTPs or hospital clusters. Zones exceeding defined concentration thresholds for priority compounds would trigger mandatory remediation plans and stricter discharge licences — directly protecting public health alongside ecosystem health.

Effective management of pharmaceutical residues in freshwater systems will reduce chronic toxicological pressure across all trophic levels — from algal primary producers through invertebrate consumers to fish — preserving ecosystem function and biodiversity.

References

Klanovicz, N., Scapini, T., Dalastra, C., de Souza, L.P., de Jesus, J.M.S., Treichel, H. and Teixeira, A.C.S.C. (2021). Antiepileptic Drugs: From Public to Environmental Health Problem. *Biochar and its Application in Bioremediation*, pp.209–229. doi: https://doi.org/10.1007/978-981-16-4059-9_10

Giebułtowicz, J., and Nałęcz-Jawecki, G. (2014). Occurrence of antidepressant residues in the sewage-impacted Vistula and Utrata rivers and in tap water in Warsaw (Poland). *Ecotoxicology and Environmental Safety*, 104, pp. 103–109. doi: https://doi.org/10.1016/j.ecoenv.2014.02.020

Zdziobek, P., and Stanek, K. (2020). The most recently used occurring residues of antidepressants in the aquatic environment, *Conditions of Power Engineering and Environmental Protection 2020.* https://www.researchgate.net/profile/Patryk-Zdziobek/publication/349822029_The_most_recently_used_occurring_residues_of_antidepressants_in_the_aquatic_environment/links/60423a424585154e8c781d4d/The-most-recently-used-occurring-residues-of-antidepressants-in-the-aquatic-environment.pdf

Liu, Q., Wang, L., Xu, X., Yan, S. and Zhu, D. (2023). Antiepileptic drugs in aquatic environments: Occurrence, toxicity, transformation mechanisms and fate. *Critical Reviews In Environmental Science and Technology*, [online] 53(21), pp.1–25. doi: https://doi.org/10.1080/10643389.2023.2209010.

Salahinejad, A., Meuthen, D., Attaran, A., Chivers, D.P. and Ferrari, M.C.O. (2023). Effects of common antiepileptic drugs on teleost fishes, *Science of The Total Environment*, 866, p. 161324. doi: https://doi.org/10.1016/j.scitotenv.2022.161324

Bagnis, S., Fitzsimons, M.F., Snape, J., Tappin, A. and Comber, S. (2018). Processes of distribution of pharmaceuticals in surface freshwaters: implications for risk assessment. *Environmental Chemistry Letters*, 16(4), pp.1193–1216. doi: https://doi.org/10.1007/s10311-018-0742-7.

Australian Government Department of Health, Disability and Ageing. (2025). *National Return and Disposal of Unwanted Medicines (NatRUM) Program*. [online] Available at: https://www.health.gov.au/our-work/national-return-and-disposal-of-unwanted-medicines-program-natrum? [Accessed 1 Jun. 2026].

Therapeutic Goods Administration (TGA). (2022). *Safe disposal of unwanted medicines*. [online] Available at: https://www.tga.gov.au/resources/consumer-information-and-resources/medicines-safety-and-general-information-consumers/safe-disposal-unwanted-medicines? [Accessed 1 Jun. 2026].